

Identifying Causal-Effect Inference Failure with Uncertainty-Aware Models

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Story: a model makes a treatment recommendation

A hospital deploys a neural network to recommend cancer treatment.

- A new patient's profile — age, genetics, history — is unlike anyone in the training data.
- The model has seen no one like this under either treatment arm.
- The model outputs: **“Recommend treatment. Confidence: 94%.”**

The problem

Standard causal neural networks have no mechanism to say “I don't know.” They extrapolate silently and confidently into uncharted covariate space.

Course recap: CATE and the identification chain

Individual-level estimand for units with covariates $X = x$:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

Under **unconfoundedness** $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$ and **consistency**:

$$\tau(x) = \underbrace{\mathbb{E}[Y \mid T = 1, X = x]}_{\mu_1(x)} - \underbrace{\mathbb{E}[Y \mid T = 0, X = x]}_{\mu_0(x)}$$

- $\mu_1(x)$ requires observed *treated* units near x .
- $\mu_0(x)$ requires observed *untreated* units near x .
- Both require **positivity**: $0 < e(x) < 1$.

Positivity fails silently in high dimensions

Positivity assumption

$$0 < \mathbb{P}(T = 1 \mid X = x) < 1 \quad \text{for all } x \text{ in the support of } X.$$

- In low dimensions: checked with propensity score histograms.
- In high dimensions (images, EHRs, genomics): most patients are unique — positivity fails somewhere for almost everyone.
- Consequence: some $\mu_1(x)$ or $\mu_0(x)$ must be **extrapolated**, not estimated.
- **Standard causal neural networks do not flag when they are extrapolating.**

Classical fix: propensity score trimming

- 1 Estimate $\hat{e}(x) = \hat{\mathbb{P}}(T = 1 \mid X = x)$.
- 2 **Trim:** remove units where $\hat{e}(x) < \eta$ or $\hat{e}(x) > 1 - \eta$.
- 3 Estimate CATE on the remaining data.

Intuition: units near $e(x) = 0$ or $e(x) = 1$ have few counterparts in the opposite arm — their CATE estimates are unreliable.

The fundamental limit

The propensity score captures *marginal* probability $\mathbb{P}(T = 1 \mid x)$. It does not directly tell you which arm lacks data for a specific x .

When trimming fails: the CEMNIST story

Causal-effect MNIST (new benchmark):

Digit	$e(x)$	Estimable?
9	1/9	Yes
2	1.0	No
other	0.5	Yes

Digit 9: low propensity but many untreated nines $\Rightarrow$ CATE estimable.

Digit 2: propensity = 1, no untreated twos $\Rightarrow$ CATE not estimable.

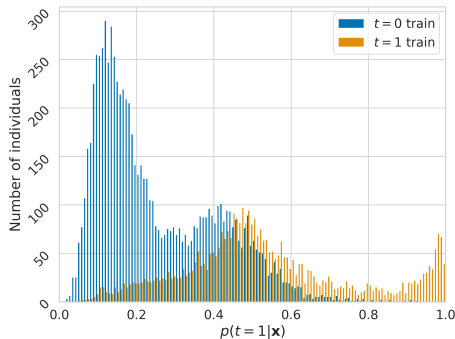


Fig. 2a: propensity histogram. Peaks at left = untreated nines. Trimmer with $\eta=0.05$ removes nines, keeps twos.

Core insight: non-overlap is covariate shift

Recall: covariate shift = test distribution differs from training distribution.

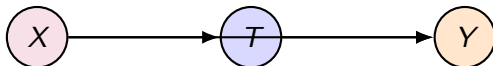
Consider the model for the $T = 1$ arm, trained on $p(X \mid T = 1)$.

Key reframing

If $e(x) = \mathbb{P}(T = 1 \mid x) \approx 0$, then by Bayes: $p(x \mid T = 1) \approx 0$.

So x is **out-of-distribution** for the $T = 1$ model.

- $e(x) \approx 0$: $\mu_1(x)$ is OoD.
- $e(x) \approx 1$: $\mu_0(x)$ is OoD.
- Non-overlap $\Leftrightarrow$ at least one outcome model must extrapolate.



Both μ_0 and μ_1 must be estimated; positivity ensures both have data.

Epistemic vs. aleatoric uncertainty

Aleatoric

Inherent noise in Y .

Cannot be reduced with more data.

Epistemic

Uncertainty from *lack of data*.

Disappears as more relevant data arrives.

We want to detect **epistemic** uncertainty in $\hat{\tau}(x)$.

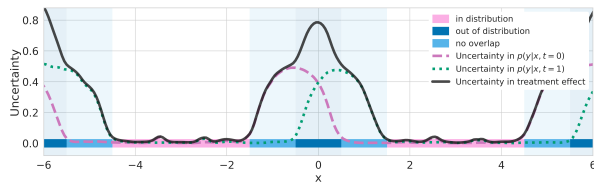
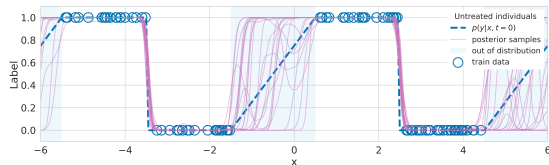
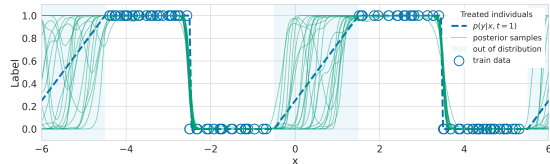


Fig. 1 (bottom): CATE uncertainty is high where at least one arm model lacks data.

Figure 1: BNN functions detect non-overlap



$T = 0$ arm: BNN functions agree in-distribution, diverge where $p(x | T = 0) \approx 0$ (right of shaded region).



$T = 1$ arm: functions diverge where $p(x | T = 1) \approx 0$ (left of shaded region = low-propensity units).

High variance across BNN draws = epistemic uncertainty = model says "I don't know."

Bayesian neural networks: the tool

Replace a single network $\mu_0(x; \omega_0)$ with a *distribution over networks*:

$$p(\omega_0 \mid \mathcal{D}) \quad (\text{posterior over weights}).$$

Epistemic uncertainty $\approx$ variance across sampled networks:

$$U_0(x) = \text{Var}_{\omega_0 \sim p(\omega_0 \mid \mathcal{D})} [\mu_0(x; \omega_0)].$$

Practical approximation: MC Dropout (Gal & Ghahramani, 2016)

Add dropout to each outcome head. Keep dropout *on* at test time. Each stochastic forward pass $\approx$ one draw from the approximate posterior.

CATE uncertainty: the formal decomposition

By the law of total variance:

$$\text{Var}[Y(1) - Y(0) \mid x] = \underbrace{\mathbb{E}_{\omega}[\text{Var}_{\gamma}[\cdots]]}_{\text{aleatoric}} + \underbrace{\text{Var}_{\omega}[\mu_{\omega_1}(x) - \mu_{\omega_0}(x)]}_{\text{epistemic in } \hat{\tau}(x)}$$

The paper uses the epistemic term as its uncertainty signal:

$$U(x) = \text{Var}_{\omega}[\hat{\tau}(x)] = \text{Var}_{\omega}[\mu_{\omega_1}(x) - \mu_{\omega_0}(x)].$$

- Estimated by sampling ω via MC Dropout at test time.
- $U(x)$ is large when at least one arm model is OoD at x .
- Captures both non-overlap *and* input covariate shift.

Neural CATE estimators: a brief map

Model	Key idea	Bayesian extension
T-Learner	Fit μ_0, μ_1 separately	B-TLearner
TARNet	Shared representation, separate outcome heads	BTARNet
CFRNet	TARNet + MMD regularisation for overlap	BCFR-MMD
Dragonnet	TARNet + joint propensity prediction	BDragonnet
CEVAE	Latent confounders via VAE	BCEVAE + neg. sampling

Paper's recipe: add MC Dropout to the outcome heads of any model $\Rightarrow$ get $U(x)$ for free, with minimal code changes.

Rejection policy: thresholding on uncertainty

Decision rule: given rejection budget r_{rej} , withhold recommendation for the r_{rej} fraction with highest $U(x)$.

Uncertainty policy

reject x if $U(x) > \kappa$

Detects non-overlap and covariate shift.

Propensity trimming

reject x if $\hat{e}(x) < \eta$ or $> 1-\eta$

Captures marginal probability only.
Trims nines, keeps twos (CEMNIST).

Metric: recommendation error rate and $\sqrt{\text{PEHE}}$ on the non-rejected subset, at fixed r_{rej} .

Results: CEMNIST — uncertainty beats propensity

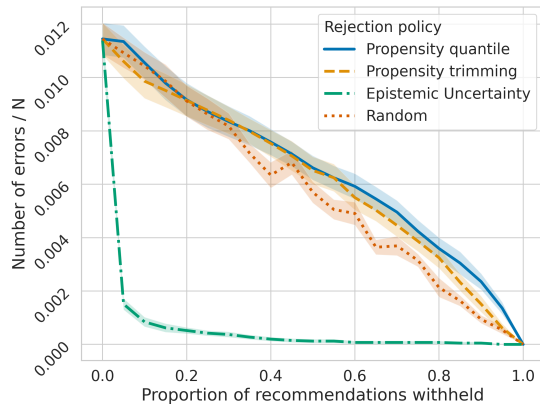


Fig. 2b: error rate vs. rejection rate on CEMNIST. **Green** (uncertainty) $\ll$ orange/blue (propensity) across the full range.

Results: IHDP and ACIC 2016

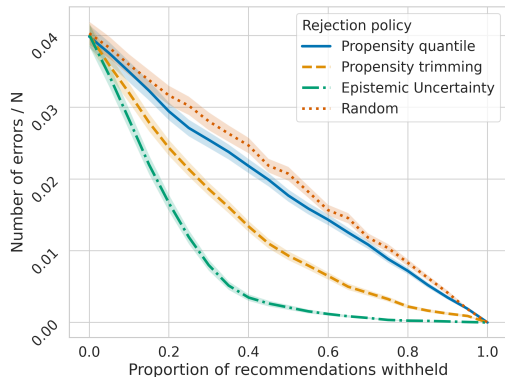


Fig. 3b: IHDP Covariate Shift. Uncertainty (green) outperforms propensity across all rejection rates.

Same result on:

- IHDP (no shift)
- ACIC 2016

All model classes improve.

Uncertainty is **more data-efficient**:
lower error at the same rejection rate.

What Jesson et al. (2020) show

- 1 Non-overlap = covariate shift for one outcome model.
- 2 Epistemic uncertainty in BNNs detects this, and detects input covariate shift as a bonus.
- 3 MC Dropout is a cheap, practical approximation.
- 4 Uncertainty-based rejection outperforms propensity trimming, especially when $e(x)$ and CATE estimability are decorrelated.

Now: how much should we trust this?

Assumption audit

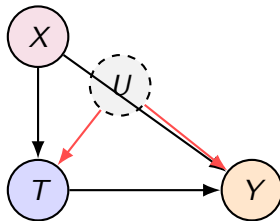
Assumption	Paper status	Notes
Unconfoundedness: $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$	Assumed, untested	Foundation of all CATE claims
SUTVA / consistency	Assumed, untested	Standard in literature
Positivity $0 < e(x) < 1$	Relaxed	The paper's contribution
Correct model class	Assumed	NN may misspecify μ_t
i.i.d. training data	Assumed	Violated under deployment shift

The paper relaxes positivity. It inherits all other classical assumptions intact.

The unconfoundedness blindspot

The method tells you **when the model is uncertain**. It does not tell you **whether the model is biased**.

If unconfoundedness fails — a hidden U confounds both T and Y :



- $\hat{\mu}_1(x)$ and $\hat{\mu}_0(x)$ both absorb the confounding signal.
- $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$ can be small even when bias is large.

A Bayesian causal model can be *confidently wrong* under hidden confounding.

Two failure modes: only one is addressed

	Low bias	High bias (hidden confounding)
High $U(x)$	Reject $\Rightarrow$ safe.	Reject $\Rightarrow$ safe (by luck).
Low $U(x)$	Recommend $\Rightarrow$ safe.	Recommend $\Rightarrow$ confidently wrong. Blind spot.

The paper operates in the left column and top row.
The bottom-right cell is untouched.

What about sensitivity analysis?

Classical tools address hidden confounding directly:

- **Rosenbaum bounds:** how large must hidden U be to overturn the conclusion?
- **E-values:** minimum confounder strength to explain away the effect.
- **Random common cause refuters** (DoWhy): add a synthetic unobserved confounder; check if the estimate changes.

The paper has no sensitivity analysis.

It tells us *when* to trust a CATE estimate (overlap, in-distribution). It does not tell us *how robust* the estimate is to hidden confounding.

Uncertainty quantification and sensitivity analysis are **orthogonal**. A complete pipeline needs both.

The benchmarking problem

All experiments use **semi-synthetic** datasets:

- **IHDP**: real covariates from a clinical trial; synthetic outcomes.
- **ACIC 2016**: synthetic competition data; known ground truth.
- **CEMNIST**: MNIST digits; synthetic treatment and outcomes.

In semi-synthetic data, true CATE is known by construction $\Rightarrow \sqrt{\text{PEHE}}$ is computable.

In real-world observational data

$Y_i(0)$ for treated units is **never observed**.

$\sqrt{\text{PEHE}}$ is **uncomputable**.

The paper's main metric cannot be verified in deployment.

Comparison to classical methods

	Propensity trimming	Uncertainty rejection
Detects non-overlap	Yes, via $\hat{e}(x)$	Yes, more precisely
Handles covariate shift	No	Yes
Handles hidden confounding	No	No
Sensitivity analysis	Rosenbaum / E-values	None
Computationally cheap	Yes	No (BNN training)
Principled threshold	Caliendo & Kopeinig	No principled rule
Verifiable on real data	Yes	Yes, but $\sqrt{\text{PEHE}}$ uncomputable

What the paper genuinely contributes

- ① **Elegant reframing.** Non-overlap = covariate shift for one arm. Opens a toolkit (BNNs, OoD detection) not previously applied here.
- ② **Practical method.** MC Dropout adds uncertainty to any neural causal model with minimal engineering cost.
- ③ **New benchmark.** CEMNIST exposes a concrete failure mode of propensity trimming in high-dimensional, reproducible settings.
- ④ **Better rejection.** Outperforms propensity trimming on IHDP, ACIC, and CEMNIST across five model architectures.

What it does not contribute

A solution to hidden confounding, a sensitivity analysis, or a way to validate CATE estimates where ground truth is unavailable.

Discussion questions — Part A

- 1 **Uncertainty and confounding.** The paper assumes unconfoundedness throughout. If there are hidden confounders, can high $U(x)$ protect you? What would need to change to also catch confounding bias?
- 2 **Real-world validation.** The results are on semi-synthetic data. What experiment — using only observational data without known CATE — would convince you this holds in clinical practice?
- 3 **Redundancy with CFRNet.** CFRNet already penalises for overlap via MMD regularisation. Does MC Dropout on top give something genuinely new, or is it partially redundant?

- ④ **What has the model contributed?** You run the method and get high $U(x)$ for a patient. You withhold the recommendation and defer to a clinician. What has the causal model actually contributed? Is detecting uncertainty enough?
- ⑤ **Uncertainty about what, exactly?** The paper estimates $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$. Is this the same as uncertainty about the true causal effect $Y(1) - Y(0)$? What components of total uncertainty are missing?

$$U(x) = \text{Var}_\omega [\hat{\mu}_{\omega_1}(x) - \hat{\mu}_{\omega_0}(x)]$$

- **What the paper gives you:** a principled signal for when to withhold a CATE estimate — detecting non-overlap and covariate shift simultaneously.
- **What it assumes and does not question:** unconfoundedness, correct model class, i.i.d. data. High $U(x)$ is not a guard against confounding bias.

The deeper lesson

Identification and uncertainty quantification are orthogonal.

You can know exactly how uncertain you are about the wrong estimand.
Epistemic uncertainty $\neq$ robustness to assumption violations.